

Introduction to Stata

A reproducible analysis, one step at a time

ECBS5241 Central European University
Ulrich Wohak 14 September 2026

Welcome and course information

Your instructor

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Research interests

How managerial behaviour and firm dynamics shape macroeconomic outcomes.

Empirical industrial organization, applied microeconometrics,
and unstructured (text) data.

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How a teaching session works

We build the project together, starting with an empty folder.

1. **A short explanation and live coding**

Predict the result, run the command, and inspect the output.

2. **A guided exercise**

Apply the idea, investigate errors, and revise your attempt.

3. **Individual practice**

Ungraded today. Graded checkpoints in Sessions 2–5.

Bring: a laptop with Stata 18 or newer. I will supply the data.

No programming prerequisite. We create the course folder in class.

Assignments and assessment

40%

Four individual checkpoints

Sessions 2–5. Each is worth 10% of the course score.

Apply a variation of the ideas practised in class.

Pass/Fail: a total of 60/100 or higher passes.

Session 1 is ungraded. The exam meeting has no checkpoint.
Full checkpoint credit can reflect a complete, documented attempt.

60%

Final in-class exam

Session 6, on 19 October.

Covers the material taught in Sessions 1–5.

Help and AI assistance

In-class checkpoints

Use Stata's help and the course materials.

No generative AI or AI coding tools unless the activity explicitly permits them.

Final exam

Permitted resources and AI rules will be announced before the exam.

Practice and checkpoint permissions do not establish exam permissions.

You must be able to explain every submitted command and conclusion.

What you will be able to do

1. **Organize and rerun a project**

Keep original data, instructions, and results in a clear structure.

2. **Prepare data with checks**

Inspect, clean, combine, and reshape observations.

3. **Produce and explain an analysis**

Automate repetition and explain descriptive tables and graphs.

4. **Work with unfamiliar commands**

Read help and output, test your understanding, and explain your decisions.

One project we build together

Starting point: an empty Introduction-to-Stata folder.

- | | | |
|-----------|----------------|---|
| 01 | Workflow | Create folders, import data, and save our first code |
| 02 | Cleaning | Add checks and documented transformations |
| 03 | Combining data | Append, merge, reshape, and aggregate |
| 04 | Programming | Automate steps with macros, loops, and programs |
| 05 | Descriptives | Generate tables and graphs from code |
| 06 | Final exam | Individual assessment of the material from Sessions 1–5 |

By the end: a documented, reproducible project.

Five teaching sessions and a final exam. Each meeting lasts 100 minutes.

Course scope

What we practise

- Stata commands and do-files
- Data management and validation
- Descriptive statistics and graphics
- Documentation and interpretation

We will type commands inside Stata.

I will supply the data as we need them. You build your own project folder.

Outside the curriculum

- Git and version-control workflows
- The operating-system terminal
- Python, R, and other languages

What is Stata?

Software for preparing data, doing statistics,
and producing tables and graphs.

A dataset has a structure

Row	hotel_id	price
1	21894	81
2	21897	81
3	21901	85

Rows are observations.
Columns are variables.

The Stata workspace

Data Editor	Inspect the dataset
Command	Enter an instruction
Results	Read what Stata returns
Do-file Editor	Save your instructions

A reproducible analysis pipeline

A saved sequence of instructions takes original data to results.

Visible steps	Readable commands and comments record what you did and why.
Repeatable work	Rerun the same checks and calculations on updated data.
Checking and sharing	Trace a result back to its inputs and let someone else check your work.

Manual edits, copied cells, and formulas spread across a workbook can make the steps hard to retrace.

Stata is our tool for practising this approach.

Files, folders, and paths

A structure we will build gradually

```
Introduction-to-Stata/  
  data/  
    raw/          original CSVs  
    derived/      generated data  
  lectures/      do-files  
  scripts/       build scripts  
  output/        logs, tables,  
                 figures
```

A path tells Stata where a file is:

```
data/raw/hotelbookingdata-vienna.csv
```

Saving a file does not create its parent folders. We create those when needed.

A **folder** contains files and other folders.

The **course root** is the top folder shown here.

Set it as Stata's **working directory**: the starting point for relative paths.

Commands, do-files, and logs

`summarize price, detail`

`summarize` The command: calculate summary statistics

`price` The variable to summarize

`, detail` An option: request additional statistics

Do-file (.do) Saved commands, run in order

Log (.log) A record of commands and results

Dataset (.dta) Data saved in Stata's format

Reading a help entry

help summarize

A shortened reading of the syntax

`summarize [varlist] [, detail]`

- `varlist` means variable names.
- Square brackets mark optional parts.
- Omit the brackets when typing.

Where to look

Description	What does it do?
Syntax	How do I write it?
Options	What can I change?
Examples	What can I try?

Today's small question: how would you request the median price?

Today's starting point

An empty folder: Introduction-to-Stata

1. Create the first subfolders and our own do-file together.
2. Import the supplied Vienna hotel CSV and inspect the records.
3. Try saving into a missing folder, read the error, and create the folder.

We add each piece when we need it.

Today we distinguish data in memory from files saved on disk.

Reference: assessment schedule

Weights below are shares of the total course score.

Session	Date	Assessment	Weight
1	14 September	Ungraded practice	0%
2	21 September	Checkpoint 1	10%
3	28 September	Checkpoint 2	10%
4	5 October	Checkpoint 3	10%
5	12 October	Checkpoint 4	10%
6	19 October	Final in-class exam	60%

Pass threshold: 60/100.

Detailed exam instructions and permitted resources: to be announced.

Reference: where to look things up

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Course reference github.com/ulrichwohak/Introduction-to-Stata

In Stata `help summarize` or `help commandname`

Official documentation [Help and search](#) [Command syntax](#) [summarize](#)

Teaching data Békés and Kézdi, *Data Analysis for Business, Economics, and Policy* (2021). See the course's `DATA_LICENSE.md`.